

Directions. Complete all the problems. Your solutions to the problems should be submitted to Gradescope before the indicated due date above. Please follow rules regarding Gradescope submission as described in the syllabus. If you use any outside source in your solutions, you must cite the source.

Problem 1. For each function below, first determine if it is **even**, **odd**, or **neither** on the given interval. *Without calculating the integrals*, state which Fourier coefficients (a_0, a_n, b_n) must be zero.

- a) $f(x) = x^3 - 4x$ on $[-2, 2]$
- b) $g(x) = x^2 \cos(x)$ on $[-\pi, \pi]$
- c) $h(x) = e^x$ on $[-1, 1]$
- d) $k(x) = |x| + \sin(x)$ on $[-\pi, \pi]$

Problem 2. The Classic Square Wave Let $f(x)$ be a 2π -periodic function defined on $[-\pi, \pi]$ as:

$$f(x) = \begin{cases} -1 & -\pi < x < 0 \\ 1 & 0 < x < \pi \end{cases}$$

- (a) Sketch the function $f(x)$ on the interval $[-3\pi, 3\pi]$.
- (b) Is this function even, odd, or neither?
- (c) Find the full Fourier series for $f(x)$.

Problem 3. Let $f(x) = x^2$ for this problem.

- (a) Are we able to determine a Fourier series for $f(x)$ as defined? If yes, determine the full Fourier series. If not, why?
- (b) If possible, determine the Fourier series for $f(x)$ defined on $-1 \leq x \leq 1$.
- (c) If possible, determine the Fourier series for $f(x)$ defined on $-2 \leq x \leq 2$.

Problem 4. Let $f(x) = \pi - x$ be a 2π -periodic function defined on the interval $-\pi \leq x \leq \pi$.

- (a) Determine the full Fourier series expansion for $f(x)$.
- (b) Sketch both the function on the interval $-5\pi \leq x \leq 5\pi$ and its corresponding series expansion over the interval $-5\pi \leq x \leq 5\pi$ (sketch for $n = 20$). *Note: You are allowed to use graphing software.*

Problem 5. Consider the function $f(x) = \pi - x$ defined only on the interval $[0, \pi]$.

- (a) Find the **Fourier Cosine series** for $f(x)$.
- (b) Find the **Fourier Sine series** for $f(x)$.
- (c) On two separate graphs, sketch the functions that each of these series converges to on the interval $[-3\pi, 3\pi]$. Label your graphs "Even Periodic Extension" and "Odd Periodic Extension."

Problem 6. Find the Fourier series for the function

$$f(x) = \begin{cases} 1 & 0 \leq x < \frac{\pi}{2} \\ 0 & \frac{\pi}{2} \leq x < \pi \\ 0 & -\pi \leq x < 0 \end{cases}$$

Problem 7. Find the Fourier series for the function

$$f(x) = \begin{cases} 0 & -5 \leq x < 0 \\ 3 & 0 \leq x < 5 \end{cases}$$

Problem 8. Determine the **Fourier Cosine series** for $f(x) = 3x^3 - 2x$ over $[0, 2\pi]$.

Problem 9. Determine a **Fourier Sine series** for $f(x) = 13 - 4x^2$ over $[0, 4]$.

Problem 10. The Dirichlet Conditions and Linear Spaces

Background Information

A set of functions S is a **linear space** (or vector space) if it satisfies two closure properties:

1. **Closure under Addition:** If $f(x)$ and $g(x)$ are both in S , then their sum $h(x) = f(x) + g(x)$ must also be in S .
2. **Closure under Scalar Multiplication:** If $f(x)$ is in S and c is any real-valued scalar, then $k(x) = c \cdot f(x)$ must also be in S .

Let D be the set of all functions defined on the interval $[-\pi, \pi)$ that satisfy the **Dirichlet conditions** (i.e., the function is bounded, has a finite number of discontinuities, and has a finite number of extrema).

Task

Show that the set D is **not** a linear space. To do this, you will use a specific counterexample.

Consider the functions:

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} + 2x & x \neq 0 \\ 0 & x = 0 \end{cases}$$
$$g(x) = -2x$$

Complete the following steps:

- (a) Briefly explain why $f(x)$ and $g(x)$ *individually* satisfy all the Dirichlet conditions and are therefore in the set D .
- (b) Now, analyze the sum $h(x) = f(x) + g(x)$. Show that $h(x)$ **violates** at least one of the Dirichlet conditions on $[-\pi, \pi)$.
- (c) Based on your finding in (b), which closure property does the set D fail? Conclude why this proves that D is not a linear space.